Formulas:

Energy required for complete transition/cycle:

$$\text{Energy}_{\text{dynamic}} \propto \text{Capacitive load} \times \text{Voltage}^2$$

Energy required for single transition:

$$\text{Energy}_{\text{dynamic}} \propto 1/2 \times \text{Capacitive load} \times \text{Voltage}^2$$

Power required per transistor:

$$\text{Power}_{\text{dynamic}} \propto 1/2 \times \text{Capacitive load} \times \text{Voltage}^2 \times \text{Frequency switched}$$

Example:

- Some microprocessors today are designed to have adjustable voltage, so a 15% reduction in voltage may result in a 15% reduction in frequency. What would be the impact on dynamic energy and on dynamic power?

$$\frac{\text{Energy}_{\text{new}}}{\text{Energy}_{\text{old}}} = \frac{(\text{Voltage} \times 0.85)^2}{\text{Voltage}^2} = 0.85^2 = 0.72$$

$$\frac{\text{Power}_{\text{new}}}{\text{Power}_{\text{old}}} = 0.72 \times \frac{(\text{Frequency switched} \times 0.85)}{\text{Frequency switched}} = 0.61$$

Dependability:

1. Mean Time Between Failure (MTBF): It is the average time available for a system or component to perform its normal operations between failures.

- MTBF= (SUM of operational time/total number of failures)

2. Mean Time To Repair (MTTR): It is the average time required to repair a failed component.

- MTTR= (SUM of downtime periods/ total number of failures)

Availability = MTBF/(MTBF + MTTR)

$$MTBF_{\text{system}} = \frac{MTBF_{\text{single drive}}}{\text{number of drives}}$$

MTTF ∝ 1/Failure Rate

Performance of two different computers:

$$\frac{\text{Execution time}_Y}{\text{Execution time}_X} = n$$

Since execution time is the reciprocal of performance, the following relationship holds:

$$n = \frac{\text{Execution time}_Y}{\text{Execution time}_X} = \frac{\frac{1}{\text{Performance}_Y}}{\frac{1}{\text{Performance}_X}} = \frac{\text{Performance}_X}{\text{Performance}_Y}$$

AMDAHL'S LAW

$$\text{Speedup} = \frac{\text{Performance for entire task using the enhancement when possible}}{\text{Performance for entire task without using the enhancement}}$$

$$\text{Speedup} = \frac{\text{Execution time for entire task without using the enhancement}}{\text{Execution time for entire task using the enhancement when possible}}$$

Alternatively, using **Amdahl's Law**:

Overall Speedup:

$$S = \frac{1}{(1-P) + \frac{P}{S_P}}$$

Example:

$$\text{Fraction}_{\text{enhanced}} = 0.4;\ \text{Speedup}_{\text{enhanced}} = 10;\ \text{Speedup}_{\text{overall}} = \frac{1}{0.6 + \frac{0.4}{10}} = \frac{1}{0.64} \approx 1.56$$

Effective CPI (Cycles Per Instruction) Formula:

$$CPI_{\text{effective}} = \frac{\sum(IC_i \times CPI_i)}{IC}$$

MIPS (Million Instructions Per Second) Formula:

$$MIPS = \frac{\text{Instruction Count}}{\text{Execution Time} \times 10^6}$$

$$\text{CPU time} = \left(\sum_{i=1}^{n} \text{IC}_i \times \text{CPI}_i\right) \times \text{Clock cycle time}$$

$$\text{CPI} = \frac{\sum_{i=1}^{n} \text{IC}_i \times \text{CPI}_i}{\text{Instruction count}} = \sum_{i=1}^{n} \frac{\text{IC}_i}{\text{Instruction count}} \times \text{CPI}_i$$

$$\text{CPU clock cycles} = \sum_{i=1}^{n} \text{IC}_i \times \text{CPI}_i$$